A Multi-Objective Metaheuristic Approach for Student–Industry Placement Allocation with Academic Supervision

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Abstract

Esta sección resumirá la motivación del trabajo, las limitaciones del problema SIPA original, la incorporación de supervisores académicos, la propuesta metaheurística multiobjetivo y los principales resultados experimentales.

Keywords: Student–Industry Placement Allocation; academic supervision; multi-objective optimization; metaheuristics; Pareto optimization; hypervolume.

1 Introduction

The allocation of students to industry placements is a relevant task in many educational contexts, especially in training programs where professional experience is part of the learning process. This process, known as *Student–Industry Placement Allocation* (SIPA), consists of assigning students to placement positions offered by companies.

This work takes as its starting point the model proposed in *An Integer Linear Programming Model Based on Competences for Student–Industry Placement Allocation* [1]. That model formulates the SIPA problem by mainly considering the relationship between students and companies, and incorporates elements such as the preferences of both parties, competence-based matching between students and placements, the student–company distance, company quotas, and possible interdependencies between placements. However, this formulation does not explicitly incorporate the academic supervision layer, even though, in real internship allocation processes, professors are responsible for monitoring and evaluating students during their stay in the company.

By not explicitly considering academic supervision, an assignment that is valid from the student–company perspective may require additional organizational decisions before it can be implemented in a real setting. In particular, once students have been assigned to placements, those placements would still need to be assigned to academic supervisors, taking into account their supervision capacity and the location of the companies. If this dimension is not integrated into the model, the final solution may be difficult to manage from an academic perspective, for instance due to an unbalanced distribution of teaching workload or a high geographical dispersion of the placements assigned to the same supervisor. For this reason, this work addresses a modified version of the SIPA problem in which academic supervisors are explicitly incorporated as a new agent in the model.

The incorporation of professors introduces new organizational criteria that must be considered together with student and company satisfaction. In particular, the extended model

considers the maximum supervision capacity of each professor, the effort associated with the geographical dispersion of supervised placements, and the workload balance among professors belonging to the same educational center. As a consequence, the resulting problem has a clearly multi-objective nature, since it combines utility objectives for students and companies with objectives related to the organizational feasibility of academic supervision.

These additional supervision-related criteria increase the complexity of the problem and reinforce the need to use approximate methods capable of obtaining high-quality solutions within reasonable computational times. In this context, this work proposes HH-LAP-SIPA, a multi-objective metaheuristic approach designed to solve the modified SIPA problem by simultaneously considering student and company utilities, as well as organizational criteria associated with academic supervision. The aim is not to obtain a single optimal assignment, but rather to explore different compromise solutions among the objectives of the problem.

The main contributions of this work are threefold. First, it formulates a modified version of the SIPA problem that incorporates academic supervision, supervision capacity, geographical dispersion, and workload balance. Second, it proposes HH-LAP-SIPA, a multi-objective metaheuristic approach to address the resulting problem. Third, it presents an experimental analysis against different reference algorithms, using hypervolume as the main multi-objective quality indicator.

2 Related Work

This section reviews the most relevant previous works to contextualize the proposed approach. First, educational assignment problems that share structural similarities with SIPA are presented, including the allocation of students to projects, groups, or supervisors. Next, works specifically addressing *Student–Industry Placement Allocation* are reviewed, with particular attention given to the competence-based model that serves as the starting point of this work. Finally, the scalability limitations associated with exact approaches are discussed, motivating the use of approximate methods to address larger instances and more complex versions of the problem.

2.1 Educational assignment problems

Assignment problems in educational contexts constitute a broad family of combinatorial problems in which students must be assigned to resources, activities, projects, groups, or supervisors while satisfying different constraints. Although these problems are not equivalent to *Student–Industry Placement Allocation*, they share relevant structural elements with it, such as the presence of students, preferences, capacities, feasibility constraints, and, in some cases, academic supervisors.

A related problem is *Student–Project Allocation* (SPA), where students must be assigned to projects while taking into account their preferences and different constraints. Several works have addressed this problem through integer programming models or approaches focused on stability, preferences, and side constraints [2, 3]. These works are relevant because they share with SIPA the need to combine preferences, capacities, and feasibility constraints, although in SPA the assigned resource is usually an academic project rather than an industry placement offered by a company.

Another group of related works focuses on the formation of student teams. In these problems, the goal is not to assign students to external entities, but to divide a class into teams satisfying certain pedagogical or organizational criteria. Recently, an evolutionary metaheuristic has been proposed for forming teams in the classroom with constraints, considering

flexible team sizes and compulsory or forbidden combinations of students [4]. Although this problem also differs from SIPA, it is relevant because it illustrates the importance of considering realistic constraints in educational assignment problems and the usefulness of metaheuristic methods when combinatorial complexity limits the applicability of exact approaches.

There are also works specifically focused on assigning students to academic supervisors. These problems consider aspects such as student and supervisor preferences, minimum and maximum supervision quotas, and workload balance among supervisors [5]. These elements are particularly close to the extension proposed in this work, since they explicitly introduce the figure of the academic supervisor. However, such approaches usually treat student–supervisor allocation as an independent problem, whereas in this work academic supervision is integrated into a broader SIPA problem that also considers students, companies, industry placements, and organizational criteria associated with supervision.

2.2 Student–Industry Placement Allocation

The *Student–Industry Placement Allocation* problem focuses on assigning students to internship placements offered by companies. Unlike other educational assignment problems, SIPA explicitly involves entities external to the educational institution, since the available placements depend on collaborating companies and on the conditions associated with each offer. Therefore, in addition to assigning each student to a suitable placement, the problem must consider constraints specific to the industry context, such as the number of available placements, the quotas agreed with each company, or the existence of placements that are related to each other.

Previous works have addressed the problem of assigning students to industry placements from specific perspectives. For example, a greedy algorithm has been proposed for a real case of student allocation in internship programs, considering the distance between the assigned professors and the locations of the internships [6]. Student placement allocation has also been studied using integer programming and *ant colony optimisation*, considering scenarios in which companies may offer several placements and express preferences over students [7]. These works show the practical relevance of the problem and the need to use optimization techniques, although they address only some of the criteria involved in real allocation scenarios.

A more directly related advance is the model proposed in *An Integer Linear Programming Model Based on Competences for Student–Industry Placement Allocation* [1]. This model formulates SIPA as a multi-objective integer linear programming problem in which both student and company utilities are considered. Its main contribution is the representation of preferences through competence vectors, relating the students' current competences, the competences they wish to develop during the placement, and the competences required by the offered placements. In addition, it incorporates the student–company distance, minimum and maximum quotas per company, and interdependencies between placements.

This competence-based model constitutes the direct starting point of the present work. However, its formulation focuses on the student–company assignment and does not explicitly incorporate academic supervision as a structural part of the problem. Therefore, the proposed extension preserves the main elements of the competence-based model, but introduces academic supervisors and organizational criteria associated with supervision, such as supervision capacity, geographical dispersion, and workload balance.

2.3 Scalability limitations and approximate methods

Exact models based on integer linear programming provide a rigorous way to formulate the SIPA problem and incorporate relevant constraints from the allocation context, although their practical applicability may be limited as instance size increases. In the competence-based SIPA model, the number of variables and constraints grows with the number of students, placements, and companies, as well as with the presence of quotas and interdependencies between placements. As a consequence, exact resolution may require high computational times, especially in scenarios with a larger number of students or with more available placements than students.

This limitation has been experimentally analysed in previous work on the SIPA ILP model, where the resolution time was observed to increase significantly as the number of students and the number of available placements grew [8]. Moreover, although commercial solvers may provide better performance than non-commercial alternatives, they are not always available to institutions that require open and accessible solutions. These results show that, although exact approaches are suitable for formally defining and validating the problem, they may not be sufficient to address larger instances within reasonable computational times.

The need for approximate methods becomes even clearer in the modified version of the problem considered in this work. The incorporation of academic supervisors adds new decisions and organizational criteria, such as assigning placements to professors, supervision capacities, geographical dispersion, and workload balance. These elements increase the search space and make the exact resolution of the complete problem even more difficult. Therefore, it is necessary to study metaheuristic methods capable of generating feasible and high-quality solutions within reasonable computational times, especially in scenarios where different trade-offs among objectives must be explored.

3 Problem Formulation

The problem addressed in this work is formulated in this section. First, the original competence-based SIPA model, which constitutes the starting point of the proposal, is summarized. Then, the extension introduced through the incorporation of academic supervisors, supervision capacities, and organizational criteria related to geographical dispersion and workload balance is described. Finally, the constraints of the modified problem and its multi-objective formulation are presented.

3.1 Original SIPA model

The original competence-based SIPA model constitutes the starting point of the formulation proposed in this work [1]. In this model, an educational institution must assign a set of students to a set of internship placements offered by companies. Let S denote the set of students, P the set of available placements, and B the set of companies offering those placements. Each placement $j \in P$ belongs to a company $k \in B$, and $P_k \subseteq P$ denotes the set of placements offered by company k.

Each company may agree with the educational institution on a minimum quota q_k and a maximum quota Q_k of assigned students. Moreover, some placements may present interdependencies, for instance when several placements are part of the same company project and must be filled jointly. To represent this situation, a set $\mathcal{I}$ of interdependent placement groups is considered, where each group $I \in \mathcal{I}$ consists of a subset of placements that must be activated jointly.

The model also incorporates competence-based information. Let C denote the set of competences relevant to the allocation process, with $c = |C|$. Each placement $j \in P$ is associated with a competence relevance vector

$$\Phi_j = \langle \phi_{j1}, \ldots, \phi_{jc} \rangle,$$

where ϕ_{jl} represents the importance of competence $l \in C$ for carrying out placement j. Similarly, each student $i \in S$ is associated with two vectors: a current competence vector

$$\Psi_i = \langle \psi_{i1}, \ldots, \psi_{ic} \rangle,$$

where ψ_{il} represents the current level of student i in competence l, and a desired competence vector

$$\Theta_i = \langle \theta_{i1}, \ldots, \theta_{ic} \rangle,$$

where θ_{il} represents the interest of student i in developing competence l during the placement.

All competence vectors are assumed to be normalized so that their components belong to the interval $[0, 1]$. In addition, placement competence relevance vectors are assumed to be normalized consistently, so that the competence matching terms $\Phi_j \cdot \Theta_i$ and $\Phi_j \cdot \Psi_i$ also lie in $[0, 1]$.

Based on this information, the model defines two main utilities. On the one hand, the utility that placement j provides to student i, denoted by τ_{ij}, combines the student–company distance with the match between the competences that the student wishes to develop and the competences associated with the placement. In the original model, this utility is defined as

$$\tau_{ij} = w_i^d(1 - d_{ij}) + w_i^c(\Phi_j \cdot \Theta_i),$$

where d_{ij} represents the normalized distance between student i and the location of placement j, while w_i^d and w_i^c represent the relative importance assigned by student i to distance and competence matching, respectively, with

$$w_i^d + w_i^c = 1, \qquad w_i^d, w_i^c \in [0, 1].$$

On the other hand, the utility that student i provides to placement j, denoted by η_{ij}, measures the match between the student's current competences and the competences required by the placement:

$$\eta_{ij} = \Phi_j \cdot \Psi_i.$$

The basic decision variable of the original model is

$$x_{ij} = \begin{cases} 1, & \text{if student } i \text{ is assigned to placement } j, \\ 0, & \text{otherwise.} \end{cases}$$

Using this variable, the model aims to simultaneously maximize the aggregate utility of students and the aggregate utility of companies, ensuring that all students are assigned, that no placement is occupied by more than one student, and that placement interdependencies and the minimum and maximum company quotas are respected.

3.2 Extension with academic supervisors

The proposed modification incorporates academic supervisors into the SIPA formulation in order to capture organizational and logistical aspects that are not represented when only

students and companies are considered. In real internship allocation processes, academic supervisors are responsible for monitoring and evaluating students, and therefore the feasibility of an allocation also depends on how supervision tasks are distributed.

In the modified model, professors are introduced as a third agent, and their inclusion affects several components of the original SIPA formulation. Specifically, the set of represented agents is extended, supervision-related distance and dispersion costs are introduced, supervision capacity constraints are added, and the objective structure is expanded to include academic and organizational criteria. Thus, the proposed formulation should be regarded as a structural modification of the original SIPA model.

The original SIPA model components are assumed to remain valid unless explicitly modified. The extension presented below introduces the additional sets, parameters, decision variable, and derived quantities required to model academic supervision. The corresponding constraints and objective components are presented in the following subsections.

Let T denote the set of professors, and let $\mathcal{E}$ denote the set of educational centers. Moreover, each professor $t \in T$ belongs to exactly one educational center, denoted by:

$$e(t) \in \mathcal{E}.$$

For each center $e \in \mathcal{E}$, define the subset of its professors:

$$T_e = \{t \in T \mid e(t) = e\}.$$

The mapping $e : T \to \mathcal{E}$ is a fixed input parameter of the model and is not part of the decision process.

For each professor $t \in T$, let C_t be the supervision capacity of professor t, defined as the maximum number of internships that they can supervise according to their available teaching hours. It is assumed that $C_t > 0$ for every professor $t \in T$, and that each educational center has at least one professor, i.e., $|T_e| > 0$ for all $e \in \mathcal{E}$.

In addition, the instance is assumed to be structurally feasible. In particular, the number of available placements and the total supervision capacity must be sufficient to assign all students, and company quotas and placement interdependencies must be mutually compatible. Therefore, the following basic conditions are assumed to hold:

$$|P| \geq |S|, \qquad \sum_{k \in B} q_k \leq |S| \leq \sum_{k \in B} Q_k, \qquad \sum_{t \in T} C_t \geq |S|.$$

These conditions are necessary feasibility conditions, although the actual feasibility of an instance also depends on the compatibility between company quotas and interdependent placement groups.

To model supervision-related spatial dispersion and travel effort, we assume a precomputed and normalized distance matrix. Let d_{uv} denote the normalized distance between two locations u and v, used to model supervision effort. Distances are assumed to be given by a precomputed distance matrix, where each location belongs to the set of relevant points composed of professors' educational centers and internship locations. In particular,

$$u, v \in \{o_t \mid t \in T\} \cup P,$$

where o_t denotes the geographic location of the educational center $e(t)$ to which professor t belongs. For notation simplicity, each placement $j \in P$ is identified with its geographic location when computing supervision-related distances.

Distances are normalized using a global min–max scaling over the complete set of relevant point pairs of the instance:

$$d_{uv} = \frac{\tilde{d}_{uv} - d_{\min}}{d_{\max} - d_{\min}},$$

where $\tilde{d}_{uv}$ denotes the original non-normalized distance, and $d_{\min}$ and $d_{\max}$ denote respectively the minimum and maximum distances computed over all relevant pairs of locations. If $d_{\max} = d_{\min}$, all normalized supervision distances are set to zero.

This normalization is performed once per instance as a preprocessing step. As a result, $d_{uv} \in [0, 1]$ for all relevant pairs, ensuring that supervision-related distance terms are comparable within the instance while preserving the relative spatial structure of the problem.

The distance matrix underlying the values d_{uv} is constructed as a preprocessing step prior to solving the allocation problem, based on the available geographic information. Distances may be computed using Euclidean metrics, road network distances, or estimated travel times, depending on the desired level of modeling detail.

In the original SIPA model, the distance between a student and a placement is normalized independently for each student using a min–max scaling over the set of placements available to that student. This student-level normalization is preserved when computing the utilities τ_{ij}, as it reflects individual commuting perception.

In contrast, supervision-related distances are normalized globally at the instance level using all relevant center–placement and placement–placement pairs. This global normalization is adopted because supervision effort represents a structural and aggregated organizational cost rather than an individual preference. Consequently, individual normalization for student utilities and global normalization for supervision dispersion coexist in the model as two distinct normalization schemes.

Extended decision variable. To link students, placements, and supervisors, the original assignment variable is extended as follows:

$$x_{ijt} = \begin{cases} 1, & \text{if student } i \text{ is assigned to placement } j \text{ under} \\ & \text{the supervision of professor } t, \\ 0, & \text{otherwise,} \end{cases} \qquad \forall i \in S,\ j \in P,\ t \in T. \tag{1}$$

Aggregated supervision effort and workload balance. In order to evaluate supervision effort at the professor level, we define several derived quantities that capture the geographical dispersion of supervised placements and the workload distribution among professors.

For each professor $t \in T$, let A_t denote the set of locations that must be covered:

$$A_t = \{o_t\} \cup \left\{ j \in P : \sum_{i \in S} x_{ijt} = 1 \right\}, \qquad \forall t \in T. \tag{2}$$

Let L_t denote the length of the minimum spanning tree over the nodes in A_t, which provides a compact proxy for spatial dispersion without explicitly defining supervision routes:

$$L_t = \sum_{(u,v) \in \mathrm{MST}(A_t)} d_{uv}, \qquad \forall t \in T. \tag{3}$$

Let N_t denote the supervision load of professor t, defined as the number of supervised students or internships:

$$N_t = \sum_{i \in S} \sum_{j \in P} x_{ijt}, \qquad \forall t \in T. \tag{4}$$

For each professor $t \in T$, the normalized workload proportion is defined as:

$$p_t = \frac{N_t}{C_t}, \qquad \forall t \in T. \tag{5}$$

For each educational center $e \in \mathcal{E}$, let $\bar{p}_e$ denote the average workload proportion of its professors:

$$\bar{p}_e = \frac{1}{|T_e|} \sum_{t \in T_e} p_t, \qquad \forall e \in \mathcal{E}. \tag{6}$$

The workload imbalance within each educational center is measured through the standard deviation of workload proportions:

$$\sigma_e = \sqrt{\frac{1}{|T_e|} \sum_{t \in T_e} (p_t - \bar{p}_e)^2}, \qquad \forall e \in \mathcal{E}. \tag{7}$$

Finally, the average supervision dispersion cost per professor is defined as:

$$\bar{L}_t = \begin{cases} \dfrac{L_t}{N_t}, & \text{if } N_t > 0, \\ 0, & \text{if } N_t = 0, \end{cases} \qquad \forall t \in T. \tag{8}$$

Since $d_{uv} \in [0, 1]$ and the minimum spanning tree over $|A_t| = 1 + N_t$ nodes contains at most N_t edges, each bounded above by 1, it follows that $L_t \leq N_t$. Therefore, for $N_t > 0$, $\bar{L}_t \leq 1$, and consequently $\bar{L}_t \in [0, 1]$. When $N_t = 0$, we define $\bar{L}_t = 0$, since no supervision effort is incurred.

Under the current definitions, the inclusion of $\bar{L}_t$ and σ_e introduces non-linear dependencies, yielding a mixed-integer non-linear formulation.

3.3 Constraints

The introduction of academic supervisors requires redefining the original SIPA constraints in order to account for the extended decision variable x_{ijt}. While the conceptual role of these constraints remains unchanged, their formulation is adapted to ensure consistency with the student–placement–professor assignment structure. In what follows, constraints (R1)–(R6) correspond to adapted versions of the original SIPA model, whereas constraint (R7) arises specifically from the incorporation of professors.

For the interdependency constraints, an auxiliary binary variable $\delta_I \in \{0, 1\}$ is used for each group $I \in \mathcal{I}$, indicating whether the corresponding group of placements is activated.

(R1) Complete assignment of students. Each student must be assigned to exactly one internship placement under the supervision of exactly one professor:

$$\sum_{j \in P} \sum_{t \in T} x_{ijt} = 1, \qquad \forall i \in S. \tag{9}$$

(R2) Placement capacity. Each internship placement can be assigned to at most one student:

$$\sum_{i \in S} \sum_{t \in T} x_{ijt} \leq 1, \qquad \forall j \in P. \tag{10}$$

(R3) Maximum company quota. At most Q_k students can be assigned to placements offered by company $k \in B$:

$$\sum_{i \in S} \sum_{j \in P_k} \sum_{t \in T} x_{ijt} \leq Q_k, \qquad \forall k \in B. \tag{11}$$

(R4) Minimum company quota. At least q_k students must be assigned to placements offered by company $k \in B$:

$$\sum_{i \in S} \sum_{j \in P_k} \sum_{t \in T} x_{ijt} \geq q_k, \qquad \forall k \in B. \tag{12}$$

(R5) Interdependent placements – upper bound. Placements belonging to the same interdependency group must be assigned jointly:

$$\sum_{i \in S} \sum_{j \in I} \sum_{t \in T} x_{ijt} \leq |I|\, \delta_I, \qquad \forall I \in \mathcal{I}. \tag{13}$$

(R6) Interdependent placements – lower bound. If an interdependency group is activated, all its placements must be assigned:

$$\sum_{i \in S} \sum_{j \in I} \sum_{t \in T} x_{ijt} \geq |I|\, \delta_I, \qquad \forall I \in \mathcal{I}. \tag{14}$$

(R7) Maximum supervision capacity. Each professor can supervise at most C_t students or internships:

$$\sum_{i \in S} \sum_{j \in P} x_{ijt} \leq C_t, \qquad \forall t \in T. \tag{15}$$

3.4 Multi-objective formulation

The incorporation of academic supervisors and educational centers introduces additional organizational criteria that must be considered in the optimization process. While the original SIPA objective function captures the preferences and objectives associated with students and companies, the extended model also accounts for the geographical compactness of supervision assignments and the proportional workload balance within each educational center.

Accordingly, the modified problem is formulated as a multi-objective optimization problem with four objective components:

- maximize the overall utility of students:

$$f_1 = \sum_{i \in S} \sum_{j \in P} \sum_{t \in T} \tau_{ij}\, x_{ijt};$$

- maximize the overall company-side utility:

$$f_2 = \sum_{i \in S} \sum_{j \in P} \sum_{t \in T} \eta_{ij}\, x_{ijt};$$

- minimize the total supervision dispersion cost:

$$f_3 = \sum_{t \in T} \bar{L}_t;$$

- minimize the total workload imbalance across educational centers:

$$f_4 = \sum_{e \in \mathcal{E}} \sigma_e.$$

Let $\mathcal{X}$ denote the feasible set defined by constraints (R1)–(R7). Thus, the optimization problem can be expressed as:

$$\max_{x \in \mathcal{X}} \left(f_1(x), f_2(x), -f_3(x), -f_4(x) \right), \tag{16}$$

where $\tau_{ij} \in [0,1]$ denotes the utility of assigning student i to placement j, and $\eta_{ij} \in [0,1]$ denotes the utility of placement j or its associated company when assigned student i.

When a scalar formulation is required, the four components can be combined through a weighted sum:

$$\max_{x \in \mathcal{X}} \ \lambda_1 \sum_{i \in S} \sum_{j \in P} \sum_{t \in T} \tau_{ij}\, x_{ijt} \ + \ \lambda_2 \sum_{i \in S} \sum_{j \in P} \sum_{t \in T} \eta_{ij}\, x_{ijt} \ - \ \lambda_3 \sum_{t \in T} \bar{L}_t \ - \ \lambda_4 \sum_{e \in \mathcal{E}} \sigma_e. \tag{17}$$

The weighting parameters λ_1, λ_2, λ_3 and λ_4 control the relative importance of the four objective components and are required to satisfy:

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1, \qquad \lambda_1, \lambda_2, \lambda_3, \lambda_4 \in [0,1]. \tag{18}$$

Since the four objective components are aggregated over different sets, their numerical scales may differ across instances. Therefore, whenever the weighted formulation is used for comparison or auxiliary scalarization, the objective components should be normalized consistently.

In this work, the problem is addressed from a Pareto-based multi-objective perspective. Therefore, the weighted formulation is not used as the main optimization criterion, but it may be used as an auxiliary scalarization mechanism when required by specific procedures or analyses.

4 Proposed Method: HH-LAP-SIPA

Esta sección presentará el enfoque metaheurístico propuesto para resolver el problema modificado. El método se describirá como una propuesta algorítmica general con dos variantes.

4.1 General overview

Aquí se explicará qué significa HH-LAP-SIPA, cuál es su objetivo, por qué se propone para este problema y cuáles son sus componentes generales: generación factible, mecanismos de reparación, evaluación multiobjetivo, archivo Pareto, procedimiento local de asignación y variantes Lite/Intensive.

4.2 Solution representation

Esta subsección describirá cómo se representa internamente una solución, incluyendo asignaciones estudiante–plaza, asignaciones plaza–profesor, cargas de empresa, cargas de profesor y plazas ocupadas.

4.3 Feasible solution generation and repair

Esta subsección explicará cómo se generan y mantienen soluciones factibles durante el proceso de búsqueda. Se describirá la construcción inicial de soluciones, incluyendo la selección de plazas, el tratamiento de interdependencias, el cumplimiento de cuotas de empresa y la asignación de supervisores académicos a las plazas ocupadas, respetando también sus capacidades de supervisión. También se presentarán los mecanismos de reparación utilizados para recuperar o preservar la factibilidad cuando una solución se modifica.

4.4 Multi-objective evaluation and Pareto archive

Esta subsección explicará cómo se evalúan las soluciones según los cuatro objetivos, cómo se aplica la dominancia de Pareto y cómo se mantiene el archivo externo de soluciones no dominadas.

4.5 Local assignment procedure

Esta subsección describirá el procedimiento local de mejora o asignación utilizado por HH-LAP-SIPA para refinar soluciones factibles (el LAP), solo empleado en la asignación estudiante–práctica.

4.6 Algorithm variants

Esta subsección introducirá las dos variantes del método propuesto. Ambas comparten la misma estructura general: selección factible de un subconjunto de prácticas, resolución de dos LAP estudiante–práctica orientados a estudiantes y empresas, evaluación multiobjetivo y actualización del Pareto externo. La diferencia principal entre ambas variantes se encuentra en la capa de asignación práctica–profesor.

La comparación entre HH-LAP-SIPA-Lite y HH-LAP-SIPA-Intensive permitirá analizar si una construcción más ligera de dicha capa favorece una mayor exploración, o si una construcción más intensiva permite obtener mejores configuraciones de supervisión bajo el mismo presupuesto temporal.

4.6.1 HH-LAP-SIPA-Lite

Esta subsubsección describirá la versión ligera del algoritmo. En esta variante, para cada subconjunto factible de prácticas, se construye una única asignación práctica–profesor mediante una construcción guiada práctica a práctica. Esta asignación se reutiliza después en las dos soluciones completas generadas mediante los LAP que maximizan respectivamente la utilidad de estudiantes y la utilidad de empresas. Su menor coste computacional por iteración puede permitir explorar un mayor número de subconjuntos de prácticas bajo un presupuesto temporal fijo.

4.6.2 HH-LAP-SIPA-Intensive

Esta subsubsección describirá la versión intensificada del algoritmo. En esta variante, para cada subconjunto factible de prácticas, se generan varias asignaciones práctica–profesor mediante cargas objetivo de los profesores, agrupación espacial de prácticas y path relinking entre una asignación inicial y una asignación guía. Estas asignaciones se combinan con los LAP de estudiantes y empresas, generando más soluciones candidatas por iteración. Su objetivo es favorecer configuraciones de supervisión más elaboradas, especialmente en términos de dispersión geográfica y equilibrio de carga, aunque con un mayor coste computacional por iteración.

4.7 Algorithm overview

Esta subsección resumirá el flujo general de HH-LAP-SIPA mediante un pseudocódigo común para ambas variantes. El objetivo será mostrar de forma compacta cómo se integran la selección factible aleatoria de prácticas, la generación de asignaciones práctica–profesor, la resolución de los LAP estudiante–práctica, la evaluación multiobjetivo y la actualización del

Pareto externo. Las diferencias entre HH-LAP-SIPA-Lite y HH-LAP-SIPA-Intensive se representarán únicamente en la función encargada de generar las asignaciones práctica–profesor, evitando repetir dos algoritmos completos.

5 Experimental Setup

Esta sección describirá el diseño experimental utilizado para evaluar el método propuesto y compararlo frente a algoritmos baseline.

5.1 Problem instances

Esta subsección describirá las instancias sintéticas generadas y los principales parámetros utilizados para caracterizarlas, incluyendo el tamaño de las instancias, la estructura de estudiantes, empresas, plazas, profesores y centros educativos, así como las dependencias y familias experimentales consideradas.

5.2 Compared algorithms

Esta subsección enumerará y describirá los algoritmos incluidos en la comparación, como la baseline aleatoria factible, NSGA-III, GRASP, HH-LAP-SIPA-Lite y HH-LAP-SIPA-Intensive.

5.3 Execution protocol

Esta subsección definirá el protocolo experimental, incluyendo número de ejecuciones, límite de tiempo, semillas, hardware, criterios de parada y criterios de equidad entre algoritmos.

5.4 Evaluation metrics

Esta subsección definirá las métricas utilizadas para comparar los algoritmos. El hipervolumen se presentará como el principal indicador de calidad multiobjetivo, explicando su cálculo, la normalización de los objetivos, el sentido de cada objetivo y la selección del punto de referencia. También se introducirán métricas complementarias, como el tamaño del Pareto y el análisis por objetivo.

6 Results and Discussion

Esta sección presentará y discutirá los resultados experimentales. El análisis se organizará de forma progresiva: primero se evaluará el rendimiento global de los algoritmos mediante hipervolumen; después se estudiará su comportamiento en función del tamaño y de distintas familias de instancias; finalmente, se analizarán los objetivos individuales y las diferencias entre las variantes de HH-LAP-SIPA.

6.1 Overall hypervolume comparison

Esta subsección comparará los algoritmos globalmente utilizando el hipervolumen como métrica principal de calidad multiobjetivo. Se analizarán valores medios, desviaciones y rankings generales sobre el conjunto completo de instancias.

6.2 Hypervolume by instance size

Esta subsección analizará si el rendimiento de los algoritmos cambia en función del tamaño de la instancia, especialmente a medida que aumenta el número de estudiantes. Este análisis permitirá estudiar la escalabilidad relativa de los métodos comparados.

6.3 Hypervolume by instance family

Esta subsección analizará grupos específicos de instancias definidos por características relevantes del problema, como el ratio de ocupación de profesores, número de profesores, número de empresas, número de plazas, estructura de dependencias u otros factores que resulten significativos en el diseño experimental. El objetivo será estudiar si el comportamiento global de los algoritmos se mantiene o cambia en escenarios concretos.

6.4 Per-objective performance analysis

Esta subsección comparará los algoritmos objetivo por objetivo, analizando si algunos métodos son mejores maximizando la utilidad de los estudiantes, maximizando la utilidad de las empresas, minimizando la dispersión de supervisión o minimizando el desequilibrio de carga. Este análisis servirá para interpretar los compromisos internos de los frentes de Pareto.

6.5 Analysis of HH-LAP-SIPA variants

Esta subsección comparará HH-LAP-SIPA-Lite y HH-LAP-SIPA-Intensive, interpretando los resultados en términos del compromiso entre exploración, intensificación y coste computacional. Por tanto, permitirá analizar si la construcción intensificada de la capa práctica–profesor mejora los componentes de supervisión o si la versión ligera resulta más efectiva bajo presupuesto temporal fijo.

7 Conclusions and Future Work

Esta sección resumirá las principales conclusiones del paper, incluyendo la utilidad de incorporar supervisión académica, el rendimiento del método propuesto y las conclusiones obtenidas del análisis experimental.

También planteará líneas de trabajo futuro, como mejorar la variante intensificada, probar operadores adicionales, utilizar instancias reales, extender el modelo con nuevas restricciones o incorporar indicadores multiobjetivo adicionales.

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